

# Scalable Information Theoretic Evaluation of the Rank Statistics in Side-Channel Attacks

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**Abstract.** Evaluating the security of a device against side-channel attacks is a difficult task. One prominent strategy for this purpose is to characterize the distribution of the *rank* of the correct key among the different key hypotheses produced by a maximum likelihood attack, depending on the number of measured traces. In practice, evaluators can estimate some statistics of the rank that are used as security indicators—e.g., the arithmetic and geometric mean rank, the median rank, the  $\alpha$ -marginal guesswork, or the success rate of level  $L$ . Yet, a direct estimation becomes time-consuming as security levels increase.

In this work, we provide new bounds on these figures of merit in terms of the mutual information between the secret and its side-channel leakages. These bounds provide theoretical insights on the evolution of the figures of merit in terms of noise level, computational complexity (how many keys are evaluated) and data complexity (how many side-channel traces are used for the attack). To the best of our knowledge, these bounds are the first to formally characterize security guarantees that depend on the computational power of the adversary, based on a measure of their informational leakages. It follows that our results enable fast shortcut formulas for the certification laboratories, potentially enabling them to speed up the security evaluation process. We demonstrate the tightness of our bounds on both synthetic traces (in a controlled environment) and real-world traces from two popular datasets (Aisylab/AES\_HD and SMAesH).

**Keywords:** Side-Channel Analysis · Security Evaluations · Success Rate of Level  $L$  · Guessing Entropy ·  $\alpha$ -Marginal Guesswork · Mutual Information.

## 1 Introduction

Side-channel attacks are a powerful class of attacks that aim to recover the secret key used by a crypto-system using some side information obtained via physical measurements. The power consumption and electromagnetic emanations of an implementation are typical examples [KJJ99, GMO01].

Given a cryptographic device, its security level against such attacks is essentially characterized from the probability distribution of the rank of the correct key in the ranking of the different key hypotheses produced by the attack [VGS13]. The worst-case evaluation of security shall ideally leverage an optimal side-channel attack procedure in order to best exploit the side-information [HRG14].

However, it is both computationally and data intensive to estimate this distribution completely. Therefore, certifying laboratories generally evaluate some figures of merit of

the best side-channel attack instead. The most widespread figures of merit are the success rate of level<sup>1</sup>  $L$  and the guessing entropy (GE) of the key [Mas94], initially proposed as metrics in the side-channel context in [SMY09], and the  $\alpha$ -marginal guesswork [Pli00b]. These figures of merit can be seen as the cumulative distribution function, the expectation, and the quantiles of the rank, respectively. Nevertheless, Martin et al. observed that the rank of the key has a very large variance in a divide-and-conquer setting [MMOS16]. As a consequence, the relevance of the guessing entropy as a security metric alone can be questioned, and combining the median and the geometric mean of the rank appears to be a more meaningful approach. Besides, the direct estimation of these figures of merit can remain computationally intensive, especially as the security levels of the target cryptographic implementations increase.

## 1.1 State of the Art and Contributions

**Context** Mangard [Man04] first connected the probability of success of a *differential power analysis* (DPA) with the number of traces used for the attack depending on the *signal-to-noise ratio* (SNR) of the side-channel traces. Mangard’s analysis was based on Pearson’s correlation coefficient, which is limited to monovariate leakages. However, attacks based solely on correlation are suboptimal; in particular, they can miss some of the non-linear leakages. For this reason, mutual information has been introduced as a more reliable and conservative measure of side-channel leakages that is also better suited for multivariate leakages [SMY09]. Heuser et al. [HRG14] modeled the side-channel attack as an unintended communication channel of the key from the hardware to the attacker. They derived the *success exponent* (SE) of the *difference of mean* (DoM) [KJJ99], *correlation power analysis* (CPA) [BCO04] and *mutual information analysis* (MIA) [GBTP08] based on the *confusion coefficient* [GHR15]. While improving Mangard’s derivations, these attacks are still suboptimal, which may lead to a false sentiment of security. De Chérisey et al. [dCGRP19a, dCGRP19b] bridged this gap to the maximum likelihood attack performance by exploiting Fano’s inequality [Fan52] to bound the success rate (of level  $L = 1$ ) of a maximum likelihood attack using the Shannon entropy. This corresponds to a worst-case attack scenario for the defender, where it is assumed that the adversary perfectly knows the leakage model. This model can be profiled offline with a clone device, for instance using *template attacks* [CRR02], leading to the question of how accurate is the model estimation—a problem that is orthogonal to our investigations and is sometimes referred to as *leakage certification* [DSV14, BHM<sup>+</sup>19, CRBO24].

**Key Enumeration** In the divide-and-conquer setting, the side-channel attacker needs to recombine each key chunk to perform a guess of the full key. She can enumerate a given number of key hypotheses until she succeeds. This is the idea of DPA combined with computational power from the seminal work of Veyrat-Charvillon et al. [VGRS12]. This methodology is termed *key enumeration* in the side-channel literature and has been investigated thoroughly [VGRS12, MOOS15, DW17, BKM<sup>+</sup>15, PSG16, LMM<sup>+</sup>16, Gro18]. Key enumeration becomes computationally and memory intensive when the key rank increases.

**Key Rank Estimation.** A laboratory evaluator can benefit from the knowledge of the actual secret key to facilitate the estimation of the correct key rank. This methodology is called *key rank estimation* and was introduced in [VGS13]. It can be efficiently performed<sup>2</sup> even for large key using various approaches [GGP<sup>+</sup>15, MOOS15, PGS15, Gro18, DW19a, DW19b, DW21]. The MCRank probabilistic algorithm [CDS22] is another more efficient

<sup>1</sup>Often termed success rate of order  $L$ . We say “of level  $L$ ” to avoid confusion with the masking order.

<sup>2</sup>An open-source implementation of the histogram methods is available on SCALib [CB23].

(yet probabilistic) way to estimate the rank even with very large keys with possibly non-independent key chunks.

Rank estimation algorithms can be used to build the *security graphs* introduced in [VGS13, Figs. 7 & 8]. Essentially, the problem of efficient (in terms of computational complexity and memory) key rank estimation can be considered as solved given the state-of-the-art algorithm. Yet, as already mentioned, this approach is computationally and data intensive because it requires evaluating the rank of the correct key for a significant number of independent attacks, for each number of traces.

**Attack Score Vector as a Multivariate Gaussian Distribution** Rivain [Riv08] made the assumption/approximation that the *score vector* produced at the output of an attack is multivariate Gaussian. Under this assumption he obtained formulas to bound the success rate of level  $L$  for a key byte. Lomné et al. [LPR<sup>+</sup>14] extended this latter work in the setting of masked encodings. Finally, Zhang et al. [ZDF20] introduced the *GEEAA* and *PS-TH-GE* algorithms based on the same assumption to bound the guessing entropy of a full AES key. However, both the *PS-TH-GE* and *GEAA* reliabilities and relevance have been questioned in [YMO22]. We do not rely on this type of assumptions in this article.

**Distinction between GM and GE in the literature** In the side-channel literature a distinction has been introduced between two estimators GM and GE of the guessing entropy which can be confusing [RPC22]. It differentiates between two empirical *estimators* of the guessing entropy. Both quantities assume that  $N$  independent attacks are mounted. Note that it is not required to know the secret key to compute GM, while it is necessary to compute GE<sup>3</sup>. While their values can differ if the profiling is inaccurate or with a too few samples both estimators estimate the same quantity which is the guessing entropy i.e. the average rank for the optimal attack. For a full key, the value of  $M$  becomes prohibitively large (e.g.  $M = 2^{128}$ ) to compute exactly GE and GM. For this reason, instead of computing the rank directly, the method of histograms [VGS13] enables to approximate efficiently  $\text{rank}_i$ .

**Metric Based Evaluation** Our method is *metric-based*, it requires only to evaluate the leakages on a given dataset. It does not predict the rank for a given attack but rather bounds some chosen figure of merits (chosen summary statistics of the rank). The main advantage for this method is that it provides tendencies. It predicts the behavior of the attack if the noise level, the masking order, the number of traces or the enumeration power of the adversary changes. We detail these motivations:

1. **Scalability:** Many side-channel attacks fall in a divide-and-conquer framework. The key of the crypto-system is split into several key chunks and side-channel information is gathered about each key chunk independently. This is typically the case of many attacks against the AES-128 where the 16 S-boxes' outputs after the initial AddRoundKey are targeted. While the figures of merit can be easily evaluated for one key chunk, it is much more challenging to evaluate them for the full key. Fortunately, informational metrics, such as entropy or mutual information, scale easily to the whole key. Hence, bounds on the figures of merit in terms of informational metrics are very relevant since they provide scalable shortcut formulas to the evaluators.
2. **Dependency on the Parameters:** Bounds on the figures of merit provide insights on the impact of the noise, number of traces and computational power of the adversary.

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<sup>3</sup>Let  $\widehat{p_{K_i|Y_i}}$  be the posterior likelihood of the different key hypotheses for the  $i$ -th attack with side information  $y_i$  and correct key  $k_i$ . Let  $\tau_i$  be a permutation of the key space so that  $\widehat{p_{K_i|Y_i}}(\tau_i(1)|y_i) \geq \dots \geq \widehat{p_{K_i|Y_i}}(\tau_i(M)|y_i)$  where  $M$  is the size of the key-space. In particular, the rank of the correct key for the  $i$ -th attack is given by  $\text{rank}_i = \tau_i^{-1}(k_i)$ . On the one hand, the GM is defined by  $\text{GM} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^M k \widehat{p_{K_i|Y_i}}(\tau_i(k)|y_i)$ . On the other hand, the GE is defined by  $\text{GE} = \frac{1}{N} \sum_{i=1}^N \text{rank}_i$ .

For instance, they permit to analyze the trade-off between data complexity (number of traces) and computational complexity (number of keys enumerated at the end of the attack). This is in contrast with a direct evaluation process, which should then be repeated for each number of traces and computational power of the adversary.

3. **Combination with Security Proofs and Countermeasures:** The bounds on the figures of merit can be combined with formal security proofs that bound the side-channel adversaries' success when a countermeasure is implemented [DFS15, MRS22, BCG<sup>+</sup>23, MS23, BCGR24]. They can be used to assess the impact of the masking order on the adversaries' success.

De Chérisey et al. [dCGRP19a, dCGRP19b] bounded the success rate of an attack without enumeration in terms of mutual information using Fano's inequality. De Chérisey et al. [dCGRP19a, § 6] also showed how to bound the guessing entropy of the key in terms of mutual information, using Massey's inequality. Choudary et al. [CP17, TCRP21, RPC22] provided tight bounds on the guessing entropy based on Rényi entropy of order  $\frac{1}{2}$  and Shannon entropy. Note that while the order  $\frac{1}{2}$  for the Rényi entropy makes sense in terms of tightness given Arikan's inequalities [Ari96], there does not exist yet an operational shortcut formula such as *Mrs. Gerbers Lemma* for  $H_{\frac{1}{2}}$  to facilitate the evaluation for masked implementations. Note that the bounds used in this approach have been recently slightly improved in [BR24]. As pointed out by Martin et al [MMOS16] the guessing entropy alone can be a misleading figure of merit. In this work, we derive metric-based shortcut formula for the first time for the median rank and to the log guessing entropy of the key. Hence, our extension to the log-guessing entropy and the median rank is an important complementary contribution for the evaluators.

Our method features the exact same computational and memory costs as the previous metric-based approach from the state of the art. It cannot be compared in terms of tightness since it bounds other figures of merit (it does not compete but rather complements the state-of-the-art).

**Contributions** Including computational power with informational metrics has been a recurring issue for the side-channel analysis community. We lack expressions for:

1. the impact of the adversary's computational power. We bridge this gap by bounding the success rate of level  $L$  of the maximum likelihood attack and, consequently, the  $\alpha$ -marginal guesswork and the median of the rank.
2. the *geometric mean* rank of the key, which was observed to be very close to its median rank [MMOS16]. We bridge this gap by using the *log-guessing entropy* and deriving the optimal bound between the log-guessing entropy and the entropy. This provides an alternative bound on the median rank of the key.

## 1.2 Outline and Technical Overview

The main notations are presented in Subsection 1.3 together with some useful, known preliminary results.

Section 2 derives a bound on the success rate of level  $L$  of a side-channel attack in terms of mutual information using a variation of Fano's inequality tailored for the list decoding problem [Eli57]. As a byproduct, we show how it provides a bound on the  $\alpha$ -marginal guesswork, i.e., on the quantiles of the rank. As a particular case, we find a simple and explicit bound on the median rank of the key.

Section 3 derives a bound on the log-guessing entropy, i.e., the logarithm of the geometric mean rank of the key, in terms of mutual information using the variational characterization of entropy given by Gibbs' inequality.

The efficiency of our methods are assessed in Section 4 via simulations and in Section 5 on real-world measurements. Section 6 concludes by discussing the practical gain of

our methodology for evaluators, particularly with respect to the evaluation score in the Common Criteria [Joi20].

### 1.3 Notations

Throughout this article,  $M$  and  $L$  are strictly positive integers such that  $L \leq M$ . A random variable is denoted by an upper-case letter (e.g.,  $X$ ) while its realization is denoted by the corresponding lower-case letter (e.g.,  $x$ ). The support of the random variable is denoted by a calligraphic letter (e.g.,  $\mathcal{X}$ ). When  $\mathcal{X}$  is discrete, its cardinality is denoted by  $|\mathcal{X}|$ . The probability distribution of a random variable  $X$  is denoted by an upper-case  $P_X$  while its probability mass function (pmf) or probability density function (pdf) is denoted by a lower-case  $p_X$ .

The joint distribution of a couple of random variable  $(X, Y)$  is denoted by  $P_{X,Y}$  which factorizes as  $P_X P_{Y|X}$ , where  $P_{Y|X}$  characterizes the *channel* (a.k.a. Markov kernel) [BCGR24], that is, the random transformation that maps a random variable  $X$  to a random variable  $Y$ . We then write  $X \rightarrow \boxed{P_{Y|X}} \rightarrow Y$  or simply  $X \rightarrow Y$  for short. The conditional transitional probability density/mass function is denoted by  $p_{Y|X}$ . Some usual probability distributions are as follows:

- $\mathcal{U}(\mathcal{X})$  is the uniform distribution on  $\mathcal{X}$ ;
- $\mathcal{B}(p)$  is the Bernoulli distribution with parameter  $p$ ;
- $\mathcal{N}(\mu, \sigma^2)$  is the normal distribution with mean  $\mu$  and standard deviation  $\sigma$ .

In the sequel, we rely on the conditional entropy (equivocation)  $H(X|Y)$  and the mutual information  $I(X; Y)$  for two random variables  $X, Y$  as defined in [PW23]. The Kullback-Leibler divergence [PW23] between two distributions  $P, R$  is denoted by  $D_{\text{KL}}(P||R)$ . The binary Kullback-Leibler divergence  $d_{\text{KL}}(p||r) \triangleq D_{\text{KL}}(\mathcal{B}(p)||\mathcal{B}(r)) = p \log \frac{p}{r} + (1-p) \log \frac{1-p}{1-r}$  is defined as the Kullback-Leibler divergence between two Bernoulli distributions.

#### 1.3.1 Side Channel as an Unintended Communication Problem

We consider a secret key  $K = (K_1, \dots, K_r) \in \mathcal{K} \triangleq \mathcal{X}^r$  composed of  $r$  key chunks where each key chunk is composed of  $n$  bits, i.e.,  $\mathcal{X} = \mathbb{F}_{2^n}$ . For the AES-128, one has  $r = 16$  and  $n = 8$  which yield a 128-bit long key and a key space of cardinality  $M \triangleq |\mathcal{K}| = |\mathcal{X}|^r = 2^{128}$ .

We consider  $q$  uniformly distributed but publicly known (plain)-text  $T^q$  and the corresponding sensitive values  $X^q$  which are assumed i.i.d. given the secret  $K$ . We assume that the adversary obtains  $q$  leakages  $Y^q$  about the sensitive variables  $X^q$ , through the memoryless and stationary channel  $X \rightarrow \boxed{P_{Y|X}} \rightarrow Y$ . More precisely, in the divide-and-conquer setting, with a plaintext  $T = (T_1, \dots, T_r)$  we assume that the sensitive values decompose as  $X = (X_1, \dots, X_r)$  where  $X_i$  depends only on  $K_i, T_i$  and each chunk of the key leak independently, so that the overall channel factorizes as  $P_{Y|X} = \prod_{i=1}^r P_{Y_i|X_i}$ . A typical channel example is  $X_i \rightarrow Y_i = w_H(X_i) + N_i$  where  $w_H$  denotes the Hamming weight function and  $N_i$  is an additive white Gaussian noise.

The adversary processes the side information with a distinguisher  $\mathcal{D}$  to recover the secret  $K$ , as illustrated by the Markov chain

$$K \rightarrow (X^q, T^q) \rightarrow \boxed{P_{Y|X}^q, \text{Id}^q} \rightarrow (Y^q, T^q) \rightarrow \boxed{\mathcal{D}} \rightarrow Z \quad (1)$$

where  $Z$  is typically either a key *guess*  $\hat{K}$ , a *list* of  $L$  guesses by decreasing order of likelihood  $\hat{K}_L$ , or a *full* ranking (i.e., a permutation of the key space.)  $R$  and  $\text{Id}$  is the identity channel. Since the full ranking information encompasses the other cases, we shall always assume in the following that the distinguisher outputs a full ranking  $R$ <sup>4</sup>.

<sup>4</sup>Note that in practice, the side-channel analysis does not output the full ranking when the key space becomes large for computational and memory constraints.

In particular, de Chérisey et al. [dCGRP19a, dCGRP19b] proved the following linear bound on the mutual information of the adversary using  $q$  traces in terms of the “single letter” mutual information of the adversary (with one trace):

**Proposition 1** (Linear Bound).  $I(K; Y^q, T^q) \leq I(K, T^q; Y^q) \leq I(X^q; Y^q) \leq qI(X; Y)$ .

It can also be observed that the mutual information is bounded above by the entropy of  $K$ . This shows that the linear bound becomes loose when either  $q$  or  $I(X; Y)$  is large.

In our setting, we observe that the mutual information for the full key “tensorizes” for each key chunk:  $I(K; Y^q, T^q) = \sum_{i=1}^r I(K_i; Y_i^q, T_i^q)$ , so that we may apply the linear bound to each key chunk as follows:

$$I(K; Y^q, T^q) = \sum_{i=1}^r I(K_i; Y_i^q, T_i^q) \leq \sum_{i=1}^r \min\{H(K_i), qI(X_i; Y_i)\}. \quad (2)$$

Applying the inequalities in this order improves the bound when the leakages differ for the various key chunks. Contrarily to the *linear* bound from Proposition 1, this approach can be easily seen to yield a concave and piecewise linear bound on the mutual information. This significantly improves the bound for uneven leakages on the different bytes.

Throughout the paper we regularly consider the common example of Hamming weight leakages (of the variable or of its S-Box) with Gaussian noise. In this case the SNR is  $\frac{n}{4\sigma^2}$  and the corresponding mutual information is approximately

$$I(X; \text{HW}(\text{Sbox}(X)) + \sigma N) \lesssim \frac{1}{2} \log \left( 1 + \frac{n}{4\sigma^2} \right) \quad (3)$$

where  $X \sim \mathcal{U}(\mathbb{F}_2^n)$ , it can be checked numerically that for  $n = 8$  when  $\sigma \geq 0.6$  then the relative error of this approximation is lower than  $10^{-3}$ . This is an upper bound on mutual information. While being asymptotically equivalent for high noise to [BCPZ16, Section. 4.2]  $\frac{n \log e}{8\sigma^2}$ , this is always more precise especially for lower noise.

### 1.3.2 Definitions and Rationales of the Figures of Merit

We assume that the side-channel attack sorts the key hypotheses by decreasing order of likelihood. The success of the attack is characterized by the rank of the correct key in this ordering. Given the conditional probabilities  $p_{K|Y^q, T^q}(\hat{K}|Y^q, T^q)$ , the *rank* of the correct key  $K$ , denoted by  $\text{rank}(K|Y^q, T^q)$  [IUH22] is the position of  $K$  in the probability mass function  $p_{K|Y^q, T^q}(\hat{K}|Y^q, T^q)$  arranged in descending order<sup>5</sup>. We introduce the following figures of merit:

- The *success rate of level  $L$*   $\text{SR}_L$  is the probability that the rank is at most  $L$ :

$$\mathbb{P}_{s,L}(K|Y^q, T^q) \triangleq \mathbb{P}(\text{rank}(K|Y^q, T^q) \leq L) = F(L) \quad (4)$$

where  $F$  is the cumulative distribution function of  $\text{rank}(K|Y^q, T^q)$  [SMY09]. When no side-information is available, we say that there is a *blind guess* and we write  $\mathbb{P}_{s,L}(K)$ . For a uniformly distributed key,  $\mathbb{P}_{s,L}(K) = L/M$ . Operationally, an adversary that enumerates the list of at most  $L$  key hypotheses can succeed in her attack with probability at most  $\mathbb{P}_{s,L}$ .

- The *guessing entropy* [Mas94]  $\text{GE}$  (a.k.a. *guesswork* [Pli00a]) is the arithmetic mean of the rank of the key

$$G(K|Y^q, T^q) \triangleq \mathbb{E}[\text{rank}(K|Y^q, T^q)]. \quad (5)$$

We write  $G(K)$  for a blind guess. For a uniformly distributed key,  $G(K) = (M+1)/2$ . Operationally, if an adversary enumerates all the key hypothesis until the secret is found, its average running time scales as  $G(K|Y^q, T^q)$ .

<sup>5</sup>Ties are resolved randomly, which does not affect the value of the figures of merit.



- Martin et al. [MMOS16] observed that the geometric mean of the rank is a more relevant security criterion than its arithmetic mean. This motivates us to use the *log-guessing entropy* (called *ranking entropy* by Martin et al [MMOS16]) that we formally define as

$$\text{LG}(K|Y^q, T^q) \triangleq \mathbb{E}[\log \text{rank}(K|Y^q, T^q)]. \quad (6)$$

The exponential of the log-guessing entropy is the geometric mean of the rank. It is more conservative than the guessing entropy by Jensen's inequality:

$$\text{LG}(K|Y^q, T^q) = \mathbb{E}[\log \text{rank}(K|Y^q, T^q)] \leq \log \mathbb{E}[\text{rank}(K|Y^q, T^q)] = \log G(K|Y^q, T^q). \quad (7)$$

Equivalently,  $\exp \text{LG}(K|Y^q, T^q) \leq G(K|Y^q, T^q)$ , which can also be seen as a consequence of the arithmetic-geometric inequality. We write  $\text{LG}(K)$  for a blind guess. For a uniformly distributed key,  $\text{LG}(K) = (\log M!)/M \approx \log(M/e)$  by Stirling's approximation.

- Pliam [Pli00a] introduced the  $\alpha$ -marginal guesswork as a relevant security metric. It is defined as

$$L(\alpha)(K|Y^q, T^q) \triangleq \min_L \{L \mid \mathbb{P}_{s,L}(K|Y^q, T^q) \geq \alpha\} = F^{-1}(\alpha). \quad (8)$$

This corresponds to the *pseudo-inverse* of the cumulative distribution function  $F$  i.e. the quantile function. In particular,  $L(\frac{1}{2})(K|Y^q, T^q) \triangleq \text{Med}(K|Y^q, T^q)$  is a *median* of the rank. We write  $L(\alpha)(K)$  for a blind guess. For a uniformly distributed key,  $L(\alpha)(K) = \lceil \alpha M \rceil$ . Operationally, if an adversary wants to succeed in his attack with probability  $\alpha$ , she needs to enumerate up to  $L = L(\alpha)(K|Y)$  key hypotheses.

All these figures of merit can be applied at the scale of a byte of the key. For example,  $\mathbb{P}_{s,L}(K_i|Y_i^q, T_i^q)$  is the probability of success of level  $L$  to guess the  $i$ -th chunk of the key.

## 2 Success Rate of Level $L$ and $\alpha$ -marginal Guesswork

In this section, we derive bounds on the success rate of level  $L$  and  $\alpha$ -marginal guesswork in terms of equivocation (equivalently, in terms of mutual information).

### 2.1 Success Rate of Level $L$

In this subsection, we derive bounds on the probability of success of level  $L$ . First, the probability of success of level  $L$  of a  $M$ -ary secret can always be lower bounded as follows

**Proposition 2.** *Let  $K$  be a  $M$ -ary secret, then  $\mathbb{P}_{s,L}(K|Y, T) \geq \mathbb{P}_{s,L}(K) \geq L/M$ .*

*Proof.* Without loss of generality we assume that  $p_K(1) \geq p_K(2) \geq \dots \geq p_K(M)$  so that  $\mathbb{P}_{s,L}(K) = \sum_{i=1}^L p_K(i)$ . First, observe that  $\mathbb{P}_{s,L}(K) = \sum_{k=1}^L p_K(k) = \sum_{k=1}^L \mathbb{E}_{y,t} p_{K|Y,T}(k|y, t) = \mathbb{E}_{y,t} \sum_{k=1}^L p_{K|Y,T}(k|y, t) \leq \mathbb{E}_{y,t} \mathbb{P}_{s,L}(K|Y = y, T = t) = \mathbb{P}_{s,L}(K|Y, T)$ . Second, for all  $i > L$  we have  $p_K(i) \leq p_K(L) \leq \sum_{i=1}^L p_K(i)/L = \mathbb{P}_{s,L}(K)/L$ . But then  $1 = \sum_{i=1}^M p_K(i) = \mathbb{P}_{s,L}(K) + \sum_{i=L+1}^M p_K(i) \geq \mathbb{P}_{s,L}(K) + \mathbb{P}_{s,L}(K)(M-L)/L = \mathbb{P}_{s,L}(K)M/L$ .  $\square$

The next ingredients are so-called *Fano inequalities*. Ahlswede, Gács, and Körner [AGK76] used Fano's inequality for the *list decoding problem* where the decoder is allowed to test  $L$  values [Eli57]. While de Chérisey et al. [dCGRP19a, dCGRP19b] used Fano's inequality to bound the success rate of a maximum likelihood attack ( $L = 1$ ). The following generalization of Fano's inequality to list decoding seems to have never been used for side-channel attacks with enumerations (which can be seen as a type of list decoding [PPS12]).

**Lemma 1** (Fano Inequality for Success Rate of level  $L$ ). *Let  $K$  be a  $M$ -ary secret. The extension of Fano's inequality to the success rate of level  $L$  is*

$$I(K; Y^q, T^q) \geq d_{\text{KL}}(\mathbb{P}_{s,L}(K|Y^q, T^q) \| \mathbb{P}_{s,L}(K)) \geq d_{\text{KL}}(\mathbb{P}_{s,L}(K|Y^q, T^q) \| \frac{L}{M}) \quad (9)$$

where  $\mathbb{P}_{s,L}(K) = L/M$  for a uniformly distributed key.

For  $L = 1$  one recovers the classical Fano inequality which is known to be obtained from the data processing inequality [PW23, Thm. 3.12]. As explained by Sakai [Sak20, Eqn. 11] Fano's inequality can be easily generalized to the list decoding problem. The following proof (for completeness since it is known) relies on the data processing inequalities for the Kullback-Leibler divergence and mutual information. This bound on the probability of success of level  $L$  is especially interesting as it covers attacks where the adversary enumerates several key hypotheses. This is the first security bound accounting for the enumeration power of the adversary in the side channel literature.

### First Proof Based on Data Processing Inequalities

*Proof.* Consider the channel  $K \rightarrow Y^q, T^q \rightarrow \widehat{K}_L$  where  $K \in \mathcal{K}$  and  $\widehat{K}_L \in \mathcal{K}^L$ , then

$$I(K; Y^q, T^q) \stackrel{(a)}{\geq} I(K; \widehat{K}_L) \stackrel{(b)}{=} D_{\text{KL}}(P_{K, \widehat{K}_L} \| P_K P_{\widehat{K}_L}) \quad (10)$$

$$\stackrel{(c)}{\geq} D_{\text{KL}}(\mathcal{B}(\mathbb{P}_{s,L}(K|Y^q, T^q)) \| \mathcal{B}(\mathbb{P}_{s,L}(K))) \quad (11)$$

$$\stackrel{(d)}{=} d_{\text{KL}}(\mathbb{P}_{s,L}(K|Y^q, T^q) \| \mathbb{P}_{s,L}(K)) \stackrel{(e)}{\geq} d_{\text{KL}}(\mathbb{P}_{s,L}(K|Y^q, T^q) \| \frac{L}{M}) \quad (12)$$

which proves the lemma. (a) is the data processing inequality for mutual information applied to the Markov chain  $K \rightarrow Y^q, T^q \rightarrow \widehat{K}_L$ . (b) holds by definition of the mutual information. (c) is the data processing inequality for the Kullback-Leibler divergence. On the left hand side the couple  $(K, \widehat{K}_L)$  is data processed into to the Bernoulli distribution defined by  $\mathbf{1}_{K \in \widehat{K}_L}$  whose parameter is  $\mathbb{P}_{s,L}(K|Y^q, T^q)$ . On the right hand side the couple  $(K', \widehat{K}_L)$  where  $K'$  is an independent copy of  $K$  is data processed into to the Bernoulli distribution defined by  $\mathbf{1}_{K' \in \widehat{K}_L}$  whose parameter is  $\mathbb{P}_{s,L}(K)$ . (d) holds by definition of the binary divergence. (e) holds by the data processing inequality for the binary divergence since by Proposition 2,  $\mathbb{P}_{s,L}(K|Y^q, T^q) \geq \mathbb{P}_{s,L}(K) \geq \frac{L}{M}$ .  $\square$

### Second Proof Based on a Variation of the Classical Proof

*Proof.* Let  $E = \mathbf{1}_{K \in \widehat{K}_L}$ . We have  $H(K|Y^q, T^q) \leq H(K|\widehat{K}_L) = H(K, E|\widehat{K}_L) = H(E|\widehat{K}_L) + H(K|E, \widehat{K}_L) \leq H_b(\mathbb{P}_{s,L}(K|Y^q, T^q)) + \mathbb{P}_{s,L}(K|Y^q, T^q) \log(L) + (1 - \mathbb{P}_{s,L}(K|Y^q, T^q)) \log(M - L)$  where we used the DPI, the chain rule and the fact that the entropy of a  $N$ -ary random variable cannot exceed  $\log N$ . With some rewriting we obtain  $I(K; Y^q, T^q) \geq d_{\text{KL}}(\mathbb{P}_{s,L}(K|Y^q, T^q) \| \frac{L}{M})$ .  $\square$

The previous bound for  $L = 1$  [dCGRP19a] established that when the mutual information tends to zero, an adversary with a single trial could not do better than a random guess. However, this did not prove security against adversaries that enumerate *several* key hypotheses. We bridge this gap using this generalized Fano inequality for any number  $L$  of enumerated key hypotheses.

Figure 1 shows the bound from Lemma 1 for alphabet size  $M = 2^{128}$ . The bound allows to analyze the computational complexity (as measured by  $L$ ) versus data complexity (as measured by  $q$ ) of any side-channel attacks. The curves in Figure 1 have a noticeable elbow, we can distinguish two specific regimes in this respect, which aligns with the intuition:



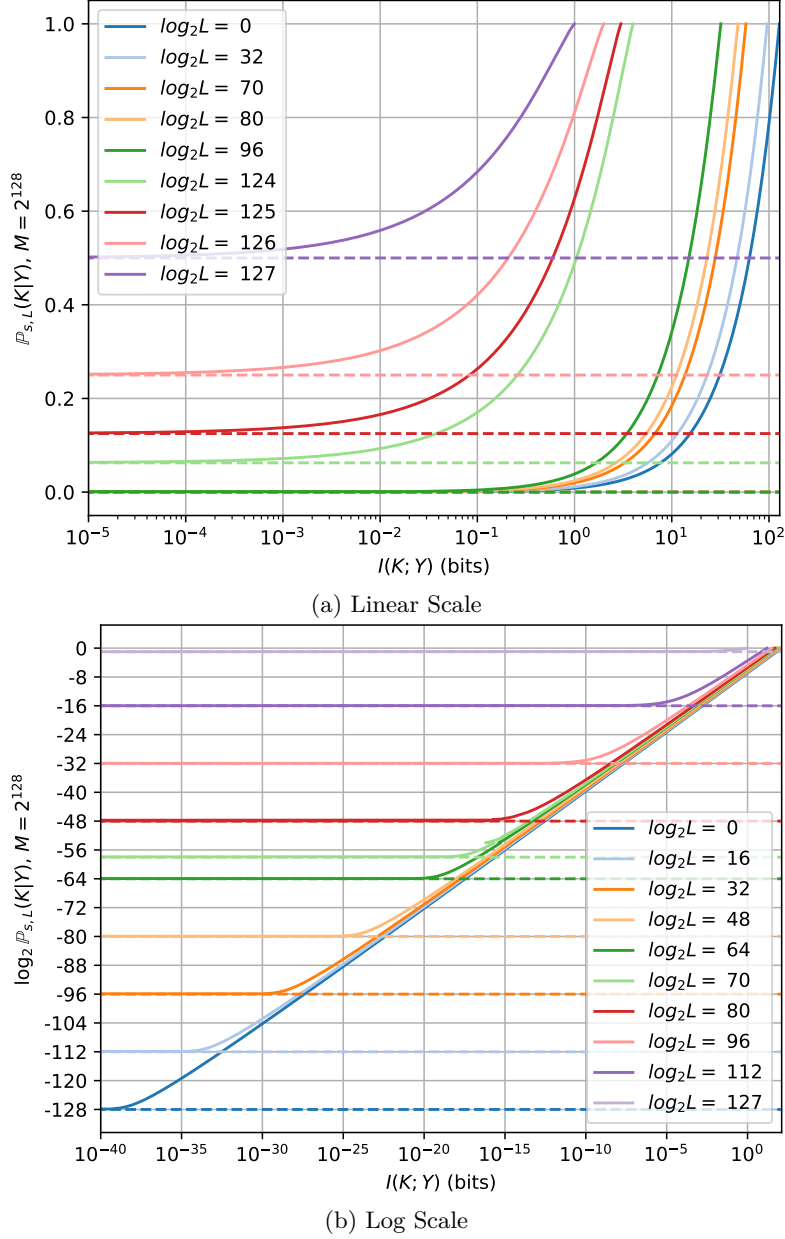


Figure 1: Fano for List Decoding, with  $M = 2^{128}$  and various values of  $L$ . The curve have a noticeable elbow shape in both the linear and log scale. The  $X$ -axis corresponds to the mutual information expressed in bits. If we assume that the linear bound is valid then this axis is proportional to the number traces. In any case the mutual information increases with the number of traces so that the impact of the number of traces can be read from the graph.

- First, accumulating more traces only affects very mildly the success rate of level  $L$  of the attack which can be seen by the plateau on the left of Figure 1b. Furthermore, the effect of the computational complexity  $L$  is linear in the probability of success of the attack. In the bound, enumerating  $L = 2^l$  key candidates amounts to apply the usual Fano's inequality with  $L = 1$  but where the size of the key space  $M = 2^{nr}$  is

replaced by an effective size of the key space  $2^{nr-l}$ . In this sense, enumerating  $2^l$  keys reduces roughly the entropy of the key by  $l$  bits which explains this factor.

- Second, the impact of the number of traces is almost linear with the probability of success. Increasing the number of traces is the best way to improve the attack performance in this regime. However, the impact of the number of enumerated key hypothesis  $L$  is notable but milder as can be observed in Figure 1a.

In the results, the most important curve is arguably for  $L = 2^{80}, 2^{96}$  which can be achieved by modern adversaries. For instance, the COPACOBANA FPGA clusters [KPP<sup>+</sup>06] from 18 years ago could enumerate  $2^{35}$  keys per second with a setup costing less than 10k dollars. In SMAesH challenge<sup>6</sup> from CHES2023, up to  $L = 2^{68}$  keys were enumerated which is the number of blocks hashed by the Bitcoin mining network in only one second.

## 2.2 Lower Bound on the Number of Traces for the Probability of Success of Level $L$

We can leverage Lemma 1 to obtain a lower bound on the minimum number of traces  $q$  needed to achieve a given success rate of level  $L$ :

**Theorem 1** (Lower Bound on the Number of Traces). *The minimum number of side channel queries  $q$  which are necessary to reduce the probability of success of level of the key  $\mathbb{P}_{s,L}(K|Y^q, T^q)$  to  $\alpha$  is lower bounded as follows:*

$$d_{\text{KL}}(\alpha \| L/M) I(X; Y)^{-1} \leq q. \quad (13)$$

Theorem 1 generalizes the lower bound on  $q$  for the success rate of level  $L = 1$  [dCGRP19a, Eqn. 15] to an arbitrary value of  $L$  hence taking into account the computational power as used in side-channel attacks with key enumeration [VGRS12]. If a countermeasure such as masking is implemented it can be combined with upper bound on  $I(X; Y)$  depending on the masking order [BCG<sup>+</sup>23, MS23].

*Proof.* By Lemma 1 with the linear bound (Proposition 1)  $qI(X; Y) \geq I(K; Y^q, T^q) \geq d_{\text{KL}}(\mathbb{P}_{s,L}(K|Y^q, T^q) \| L/M)$ .  $\square$

## 2.3 Bound on the $\alpha$ -marginal Guesswork

We first begin by a preliminary upper bound on the  $\alpha$ -marginal guesswork:

**Proposition 3.** *For any  $M$ -ary secret  $K$  (not necessarily uniform) and  $\alpha \in (0, 1)$ :*

$$L(\alpha)(K|Y, T) \leq L(\alpha)(K) \leq \lceil \alpha M \rceil. \quad (14)$$

*In particular, for any  $y, t$ ,  $L(\alpha)(K|Y = y, T = t) \leq \lceil \alpha M \rceil$  e.g.  $\text{Med}(K|Y, T) \leq \lceil M/2 \rceil$ .*

*Proof.* Let  $\alpha \in (0, 1)$ . By definition,  $L(\alpha)(K) = \min E$  where  $E = \{L \| \mathbb{P}_{s,L}(K) \geq \alpha\}$  and  $L(\alpha)(K|Y, T) = \min E_{Y,T}$  where  $E_{Y,T} = \{L \| \mathbb{P}_{s,L}(K|Y, T) \geq \alpha\}$ . Now by Proposition 2,  $\mathbb{P}_{s,L}(K|Y, T) \geq \mathbb{P}_{s,L}(K)$  so that  $E \subseteq E_{Y,T}$ . Since the minimum of a set is at most equal to the minimum of one of its subset, the inequality  $L(\alpha)(K|Y, T) \leq L(\alpha)(K)$  follows. If  $\frac{L}{M} \geq \alpha$  then by Proposition 2,  $L \in E$  and  $L(\alpha)(K) \leq L$ . This is valid for all integer  $L$  such that  $L \geq \alpha M$  i.e.  $L \geq \lceil \alpha M \rceil$ . We obtain the result by taking  $L = \lceil \alpha M \rceil$ .  $\square$

We now derive a lower bound on  $L(\alpha)(K|Y)$  which we expect to approach  $\lceil \alpha M \rceil$  for a uniform key as  $I(K; Y) \rightarrow 0$ . The  $\alpha$ -marginal guesswork  $L(\alpha)$  can be lower bounded using Lemma 1 using the following lemma:

<sup>6</sup><https://smaesh-challenge.simple-crypto.org/>

**Lemma 2.** Let  $\phi : L \mapsto [0, 1]$  be an upper bound on  $\mathbb{P}_{s,L}$  then,

$$L(\alpha) = \min\{L | \mathbb{P}_{s,L} \geq \alpha\} \geq W(\alpha) \triangleq \min\{L | \phi(L) \geq \alpha\}. \quad (15)$$

*Proof.* Since  $\mathbb{P}_{s,L} \leq \phi(L)$  for any  $\alpha \in [0, 1]$  we have the inclusion  $\{L | \mathbb{P}_{s,L} \geq \alpha\} \subseteq \{L | \phi(L) \geq \alpha\}$ . Hence, the results since the minimum on a subset of a set is always larger than the minimum on the set itself. Lemma 2 can be visually understood with the illustration of Figure 2.  $\square$

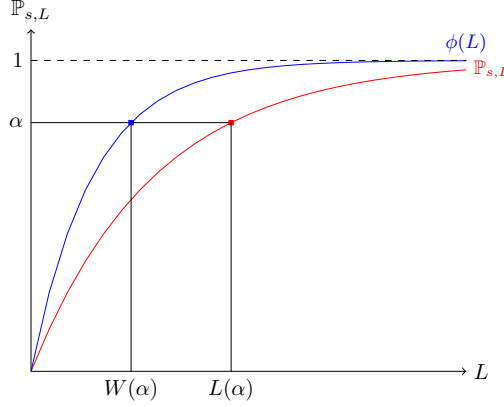


Figure 2: Visual Proof of Lemma 2.

As a consequence the upper bound on the success rate of level  $L$  (Lemma 1) yields a bound on the  $\alpha$ -marginal guesswork:

**Theorem 2.** Let  $\alpha \in (0, 1)$ ,

$$L(\alpha)(K|Y^q, T^q) \geq \min\{L | I(K; Y^q, T^q) \geq d_{\text{KL}}(\alpha \| \mathbb{P}_{s,L}(K))\}. \quad (16)$$

For the median rank of an a priori uniformly distributed key we obtain:

$$\text{Med}(K|Y^q, T^q) \geq \frac{M}{2} \left(1 - \sqrt{1 - \exp(-2I(K; Y^q, T^q))}\right) \quad (17)$$

*Proof.* We combine Lemma 1 with Lemma 2. The median is obtained for  $\alpha = \frac{1}{2}$  and  $d_{\text{KL}}(\frac{1}{2} \| \mathbb{P}_{s,L}(K)) = \frac{1}{2}(\log \frac{1}{2\mathbb{P}_{s,L}(K)} + \log \frac{1}{2(1-\mathbb{P}_{s,L}(K))}) = -\frac{1}{2} \log \frac{4L(M-L)}{M^2}$ . Hence, we have to solve the degree polynomial in  $L$ ,  $I(K; Y^q, T^q) = -\frac{1}{2} \log \frac{4L(M-L)}{M^2}$ . The discriminant is always positive so that we obtain two roots  $L = \frac{M}{2}(1 \pm \sqrt{1 - \exp(-2I(K; Y^q, T^q))})$ . Now by Proposition 3 we know that  $\text{Med}(K|Y^q, T^q) \leq M/2$  hence we can remove one of the two roots to obtain Eqn. (17).  $\square$

A bound on the median rank can be very interesting for the evaluator. It is less sensitive to large deviation than the average rank (as measured by the guessing entropy). Furthermore, it answers the following question: *How many key hypotheses should be enumerated to achieve a success rate of  $\frac{1}{2}$ ?*

A dichotomic search over  $L$  can be used to compute  $L(\alpha)$  in Theorem 2. As the leakage  $I(K; Y^q, T^q)$  approaches zero then necessarily  $\mathbb{P}_{s,L}(K)$  should approach  $\alpha$  for  $L$  to satisfy  $I(K; Y^q, T^q) \geq d_{\text{KL}}(\alpha \| \mathbb{P}_{s,L}(K))$ . In other words  $L$  should approach  $\alpha M$ . This is in line with the upper bound from Proposition 3. Theorem 2 yields the bounds from Figure 3.

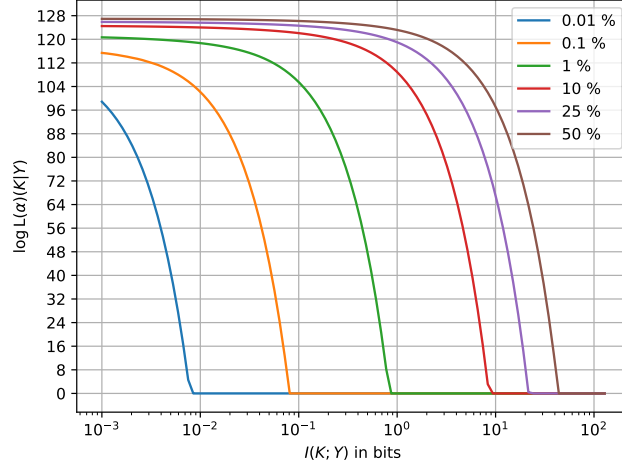


Figure 3: Illustration of Theorem 2 for a 128 bits key.

*Remark 1* (Tightness). When the mutual information becomes large Theorem 2 becomes vacuous. This happens when  $\frac{M}{2}(1 - \sqrt{1 - \exp(-2I(K; Y^q, T^q))}) \leq 1$  i.e.  $I(K; Y^q, T^q) = -\frac{1}{2} \log(\frac{4}{M}(1 - \frac{1}{M})) \approx \frac{nr-2}{2}$  bits. In other words, our bound on the median becomes vacuous when approximately half of the secret is leaked.

**Example 1** (Behavior For Hamming Weight Leakages). For Hamming weight leakages, using Eqn. (3), we have approximately

$$\text{Med}(K|Y^q, T^q) = \frac{M}{2} \left( 1 - \sqrt{1 - \left(1 + \frac{n}{4\sigma^2}\right)^{-qr}} \right) \quad (18)$$

Now if  $\sigma \rightarrow +\infty$ ,

$$\text{Med}(K|Y^q, T^q) \geq \frac{M}{2} \left( 1 - \frac{\sqrt{qrn}}{2\sigma} \right). \quad (19)$$

## 2.4 Lower Bound on the Number of Traces for the Median Rank

Leveraging Theorem 2, we lower bound the minimum number of traces  $q$  that are needed to achieve a given median rank of the key:

**Theorem 3** (Lower Bound on the Number of Traces). *The minimum number of side channel queries  $q$  which are necessary to reduce the median rank of the key  $\text{Med}(K|Y^q, T^q)$  to  $\text{Med}$  is lower bounded as follows:*

$$q \geq -\frac{1}{2I(X; Y)} \log \left( 4 \frac{\text{Med}}{M} \left( 1 - \frac{\text{Med}}{M} \right) \right). \quad (20)$$

*Proof.* Theorem 2 implies that  $\text{Med}(K|Y^q, T^q) \geq \frac{M}{2}(1 - \sqrt{1 - \exp(-2I(K; Y^q, T^q))})$ . Hence,  $1 - \exp(-2I(K; Y^q, T^q)) \geq (1 - \frac{2}{M} \text{Med}(K|Y^q, T^q))^2$ . That is  $4 \frac{\text{Med}(K|Y^q, T^q)}{M} (1 - \frac{\text{Med}(K|Y^q, T^q)}{M}) \geq \exp(-2I(K; Y^q, T^q))$ . Using the linear bound (Proposition 1),  $qI(X; Y) \geq I(K; Y^q, T^q) \geq -\frac{1}{2} \log(4 \frac{\text{Med}(K|Y^q, T^q)}{M} (1 - \frac{\text{Med}(K|Y^q, T^q)}{M}))$  which concludes the proof.  $\square$

## 3 Log Guessing Entropy or Geometric Mean Rank

The logarithm of the geometric mean rank of the key is an interesting security metric for an evaluator as it represents the average of the number security bits of the key. As

observed by Martin et al. [MMOS16] this is more reliable metric than the logarithm of the guessing entropy. This quantity is captured by the log-guessing guessing entropy. In this section, we derive both a lower and an upper bound on the log-guessing entropy in terms of equivocation (or mutual information) whose evaluation scales to the full key.

### 3.1 Lower Bound on Log-Guessing Entropy

We derive the optimal lower bound on conditional log-guessing entropy in terms of conditional entropy:

**Theorem 4.** Let  $\phi_M(a) \triangleq \ln(\sum_{i=1}^M i^{-a})$ . For all  $a \in \mathbb{R}^+$ ,

$$H(K|Y^q, T^q) \leq a\text{LG}(K|Y^q, T^q) + \phi_M(a)(\log e). \quad (21)$$

The optimal lower bound on  $\text{LG}(K|Y^q, T^q)$  in terms of  $H(K|Y^q, T^q)$  is given by a parametric curve  $\{(H(a), \text{LG}(a)) | a \in \mathbb{R}^+\}$ , where

$$\begin{aligned} H(a) &\triangleq (\phi_M(a) - a \frac{d\phi_M}{da}(a))(\log e), \\ \text{LG}(a) &\triangleq -\frac{d\phi_M}{da}(a)(\log e). \end{aligned} \quad (22)$$

*Proof.* First, observe that Eqn. (21) is linear in  $H$  and  $\text{LG}$  hence it is enough to prove it in the unconditional setting. The conditional version is obtained from the unconditional one by averaging the inequality for each  $Y^q, T^q = y^q, t^q$ .

Let  $p$  be the secret's pmf, without loss of generality we assume that  $p(1) \geq p(2) \geq \dots \geq p(M)$ . For any pmf  $r$  we have by Gibbs' inequality (Thm. 2.6.3 in [CT01]) that  $H(K) \leq -\sum_{k=1}^M p(k) \log r(k)$ . The upper bound should depend on the secret only through its log-guessing entropy  $\text{LG}(K)$ .

This is the case when  $\log r(k) = -a \log k + b$  where  $b$  is a normalization factor i.e.  $r(k) = k^{-a} (\sum_{k=1}^M k^{-a})^{-1}$  is a Zipf's law (a.k.a. discrete Pareto distribution) of parameter  $a \in \mathbb{R}^+$ . We obtain,  $H(K) \leq -\sum_{k=1}^M p(k)(-a \log k - \phi_M(a)(\log e)) = a\text{LG}(K) + \phi_M(a)(\log e)$  which proves Eqn. (21). Furthermore, Gibbs' inequality is achieved with equality when  $p = r$  for which  $\text{LG}(K|Y^q, T^q) = \text{LG}(a) = -\frac{d\phi_M}{da}(a)(\log e)$  and  $H(K|Y^q, T^q) = H(a) = (\phi_M(a) - a \frac{d\phi_M}{da}(a))(\log e)$  which proves Eqn. (22).  $\square$

When  $a = 1$ ,  $\phi_M(1) \approx \ln(H_M)$  where  $H_M = \ln M + \gamma + \frac{1}{2M} + o(\frac{1}{M})$  is the  $M$ -th harmonic number and  $\gamma \approx 0.577$  is EulerMascheroni's constant. When  $a \neq 1$  we suggest to use comparison with integral techniques such as [Nal15, Eqn. 16] to approximate tightly  $\phi_M(a)$  and its derivative to avoid the naive computation of the sum in their definition whose complexity would be linear in  $M$  which can be intractable.

This bound on the log-guessing entropy is useful for the evaluator. It is less sensitive to large deviation than the guessing entropy. In a sense it gives the average security parameter of the implementation<sup>7</sup>.

*Remark 2* (Tightness). David and Wool [DW15, § 5] experimentally observed that the distribution of the rank is long-tailed and well approximated by a Pareto distribution. Since the rank is discrete and bounded, their observation can be refined by saying that the rank is well approximated by a truncated Zipf law (which is the discrete analog of a truncated Pareto distribution). Martin et al. [MMOS16, § 4.2] reported that the rank is well approximated by a delta-log-normal distribution which is another type of "heavy tailed" distribution. As Theorem 4 is achieved with equality for a truncated Zipf law we expect our bound to be very tight on practical side-channel traces.

<sup>7</sup>while the log of the guessing entropy would be the security parameter associated to the average security

### 3.2 Upper Bound on Log-Guessing Entropy

McEliece and Yu derived an optimal upper bound on the guessing entropy in terms of entropy [MY95, Theorem 1, Eqn. 5]. We modify their derivations to obtain an upper bound on the log-guessing entropy in terms of entropy:

**Theorem 5** (Upper Bound on LG). *The following upper bound on the log-guessing entropy holds*

$$\text{LG}(K|Y^q, T^q) \leq \log(M!)(M \log M)^{-1} H(K|Y^q, T^q). \quad (23)$$

Note that  $\frac{\log M!}{M \log M} \approx 1 - \frac{\log e}{\log M}$  approaches 1 as  $M$  increases. This inequality is tight in the sense that if  $K$  is uniformly distributed and the channel  $K \rightarrow Y^q, T^q$  is an erasure channel with erasure probability  $\epsilon$  then the bound is matched everywhere as  $\epsilon$  varies from 0 to 1.

*Proof.* We slightly modify the derivations of McEliece and Yu [MY95] that derived an upper bound on the guessing entropy in terms of entropy to obtain a bound on the log-guessing entropy in terms of entropy. Let  $K_M = \text{ConvHull}(\{a_1, \dots, a_M\}) = \{p : 0 \leq p_M \leq \dots \leq p_1 \text{ and } \sum_{i=1}^M p_i = 1\}$  be a convex compact subset of the  $M$ -ary probability simplex where  $a_i = (\frac{1}{i}, \dots, \frac{1}{i}, 0, \dots, 0)$  for  $1 \leq i \leq M$  and  $f : p \in K_M \mapsto H(a_M)\text{LG}(p) - \text{LG}(a_M)H(p) \in \mathbb{R}$ . Since the entropy  $H(p)$  is concave in  $p$  and the log-guessing entropy is linear in  $p$ ,  $f$  is convex in  $p$ . As a result,  $f$  reaches its maximum value in one of the corners  $a_i$  whose convex hull spans  $K_M$  i.e.  $\max_{P \in K_M} f(P) = \max_{i=1, \dots, M} f(a_i)$ . Now observe that  $f(a_1) = 0$  and for  $i \geq 2$ ,  $f(a_i) = H(a_i)H(a_M)(\text{LG}(a_i)H(a_i)^{-1} - \text{LG}(a_M)H(a_M)^{-1}) \leq 0$  since the sequence  $\text{LG}(a_i)H(a_i)^{-1} = \log(i!)(i \log i)^{-1}$  is monotonically increasing for  $i \geq 2$ . As a consequence we have shown that if  $K$  is a  $M$ -ary random variable then  $\text{LG}(K) \leq H(K) \log(M!)(M \log M)^{-1}$ . Now, by averaging for each  $(K|Y^q = y^q, T^q = t^q)$  we obtain Eqn. (23) as desired.  $\square$

An upper bound on the log-guessing entropy in terms of entropy is useless for security evaluations as we access to upper bound on the mutual information and not lower bounds. However, it is interesting as it allows us to assess the worst case gap between our lower bound on the log-guessing entropy and its actual value. The combination of Theorem 5 with Theorem 4 (with  $a = 1$ ) yields the following sandwiching bound,

$$H(K|Y^q, T^q) - \log(H_M) \leq \text{LG}(K|Y^q, T^q) \leq H(K|Y^q, T^q) \quad (24)$$

where  $H_M = \ln M + \gamma + \frac{1}{2M} + o(\frac{1}{M})$  is the  $M$ -th harmonic number. This shows that the log-guessing entropy can be accurately bounded in terms of entropy as the sandwiching bound roughly indicates that  $\text{LG}(K|Y^q, T^q) \approx H(K|Y^q, T^q)$ . The maximal gap between the lower and upper bound is very moderate as it is approximately  $\log(\ln M)$  as  $M$  grows.

**Example 2** (Behavior For Hamming Weight Leakages). For Hamming weight leakages, using Eqn. (3), we have approximately

$$nr - \frac{qr}{2} \log(1 + \frac{n}{4\sigma^2}) - \log(nr \ln 2 + \gamma) \lesssim \text{LG}(K|Y^q, T^q) \lesssim nr - \frac{qr}{2} \log(1 + \frac{n}{4\sigma^2}). \quad (25)$$

### 3.3 Lower Bound on the Number of Traces for Log-Guessing Entropy

Combining Theorem 4 with the linear bound (Proposition 1) we obtain a lower bound on the minimum number of side channel traces needed by an adversary to reduce the log-guessing entropy of the key to a targeted value LG:

**Theorem 6** (Lower Bound on the Number of Traces). *The minimum number of side channel queries which are necessary to reduce the log-guessing entropy of the key  $\text{LG}(K|Y^q, T^q)$  to LG is lower bounded as follows:*

$$q \geq (\log(M/H_M) - \text{LG})I(X; Y)^{-1}. \quad (26)$$



## 4 Synthetic Data Evaluation

We evaluated the performance of the a maximum likelihood attack on an AES where the Hamming weight of the 16 S-boxes of the first round is leaking perturbed by additive Gaussian noise of standard deviation  $\sigma$ . We plot the result for  $\sigma \in \{1, 4, 16, 32\}$  i.e.  $\text{SNR} \in \{2, 2^{-3}, 2^{-7}, 2^{-11}\}$ . Results are given in Figure 4 and Figure 5.

We estimate the figure of merits by first estimating the ranks using the histogram method implemented in SCALib [CB23] and then using this value in the different estimators. Let  $\widehat{\text{rank}}_i$  be the estimation of the rank for the  $i$ -th attack. We estimated the guessing entropy, log guessing entropy and median rank by the respective expressions  $\frac{1}{N} \sum_{i=1}^N \widehat{\text{rank}}_i$ ,  $\frac{1}{N} \sum_{i=1}^N \log \widehat{\text{rank}}_i$  and  $\text{median}(\{\widehat{\text{rank}}_i, i = 1, \dots, N\})$ . In the simulations  $N = 1000$ , for the real dataset (in the next section)  $N$  is the minimum between 1000 and the number of available traces divided by the number of traces used to mount the attack. For the simulations, we computed the exact mutual information with a Monte-Carlo estimator using  $N_s = 10^4$  samples.

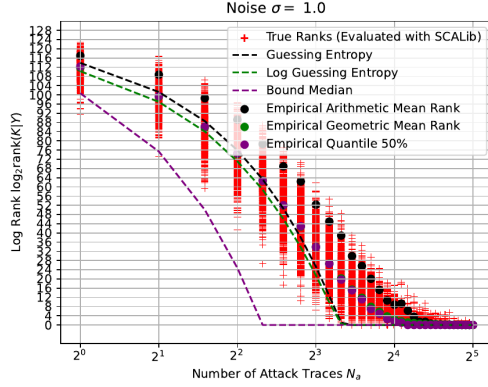
We confirm experimentally the observation of [MMOS16] that the log-guessing entropy (geometric mean rank) is very close to the median rank of the correct key in the attack. Our bound on the log-guessing entropy (Theorem 4, green dashed line) is very close to the empirical geometric mean and median rank of the key. Our bound on the median rank (Theorem 2, purple dashed line) is tight for small leakages but becomes looser as the number of traces increases. In this case, it is better to use our bound on the log-guessing entropy as a bound on the median, under the assumption that the geometric mean of the rank and its median are close.

The experiments also confirm that the arithmetic mean rank of the key can be a misleading security parameters as it is drifted above by the worst attack performance. The black dots are at the very top of the cloud of red crosses. This shows that the rank of the correct key in most attacks can be order of magnitude bellow the average rank of the correct key. For instance, with  $q = 2^{10}$  and  $\sigma = 32$  we observed that approximately 95% of the attack ranks are bellow the arithmetic mean rank for these parameters. Hence, it is more meaningful to consider the geometric mean rank of the correct key, which is the average security parameter in a sense, as a security parameter (as already advocated by Martin et al. [MMOS16]). We also plot the improved bound on the guessing entropy [BR24, Theorem 1] (black dashed line) which is close to the empirical arithmetic mean of the rank for small leakages (black dots), but the gap widens as the number of traces increases (and the bound seems actually closer to the geometric mean rank than the arithmetic mean rank).

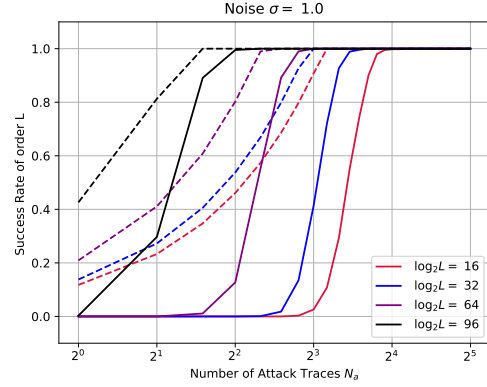
We also confirm the tightness of Lemma 1 on the success rate of level for different values of  $L$ . We observe a multiplicative gap with respect to the number of traces between the bound and the empirical values. Interestingly, this gap seems approximately constant with respect to  $L$ . We observe on Figure 4 and 5 that our bounds on the median rank (purple dashed line) and log-guessing entropy (green dashed line) are close to the actual values (purple and green dots respectively) for rank larger or equal to  $2^{40}$ . This holds in particular for the range we are interested in as an evaluator, namely range larger than  $2^{70}, 2^{808}$ . When the number of traces becomes too large the linear bound (Proposition 1) becomes looser which impacts the tightness of the final results. We attribute this gap not to our bound but to the non-tightness of the linear bound for large number of traces. In particular, for ranks lower than  $2^{40}$  our bounds on the median rank and log-guessing entropy becomes looser. However, from an evaluator perspective the implementation can be considered as broken for ranks in this order of magnitude. For this reason, the gap with our bounds does not matter since proving security for an insecure implementation does not make sense. In all our experiments (Sections 4 and 5) we estimate the figures of merits

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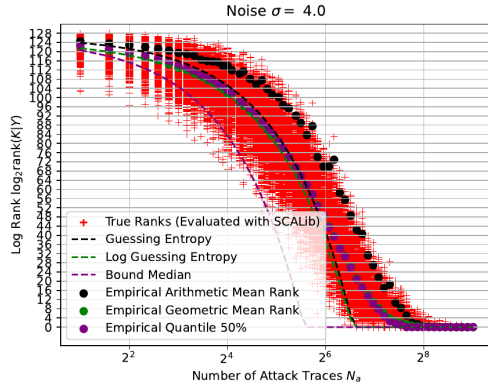
<sup>8</sup>As discussed in Section 2.1



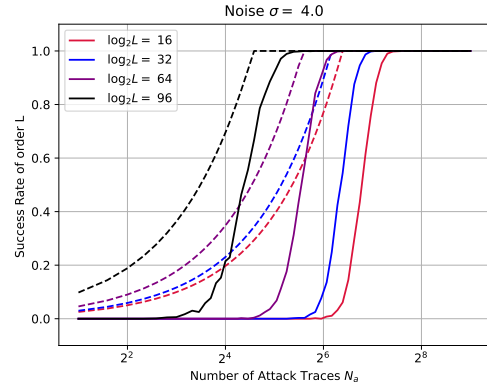
(a) Rank and  $\sigma = 1$



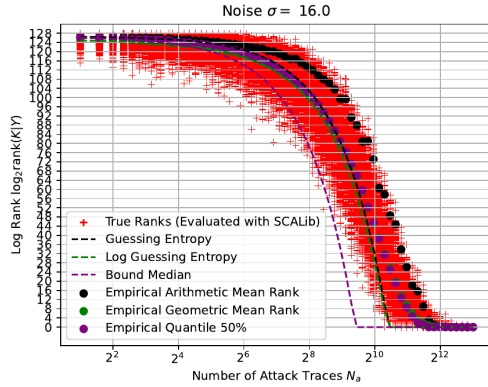
(b) Success Rate of Level  $L$  and  $\sigma = 1$



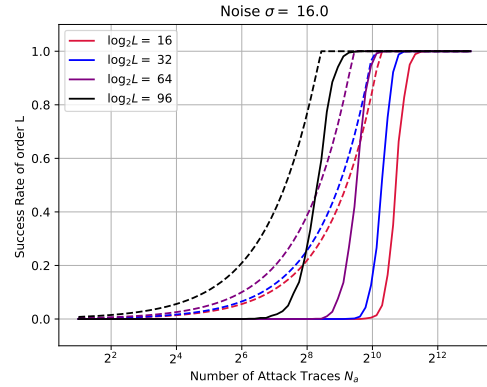
(c) Rank and  $\sigma = 4$



(d) Success Rate of Level  $L$  and  $\sigma = 4$



(e) Rank and  $\sigma = 16$



(f) Success Rate of Level  $L$  and  $\sigma = 16$

Figure 4: Success Rate of Level  $L$  and rank evaluated on 1000 attacks for different number of traces and various noise levels. The ranks evaluated using histograms on the attacks are plotted as red crosses. The dots represent the empirical values for the figures of merit while the dashed lines are the informational bounds that we derived in terms of mutual information.

by first estimating the rank using the histogram method implemented in SCALib [CB23]

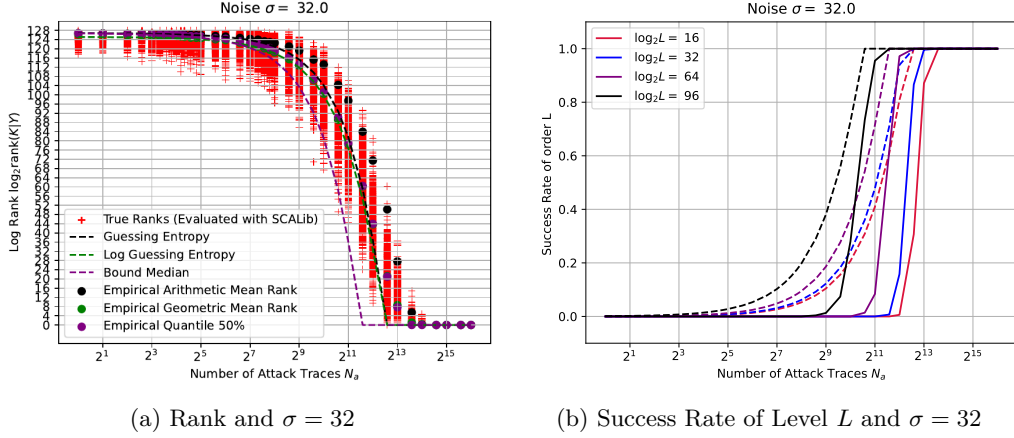


Figure 5: Success Rate of Level  $L$  and rank evaluated on 1000 attacks for different number of traces and various noise levels. The ranks evaluated using histograms on the attacks are plotted as red crosses. The dots represent the empirical values for the figures of merit while the dashed lines are the informational bounds that we derived in terms of mutual information.

and then using this value in the different estimator.

## 5 Real Data Evaluation

**Experimental Setup and Inaccurate Mutual Information.** We profiled a Regression-Based Linear Discriminant Analysis (RLDA) template using SCALib and evaluated the corresponding mutual information as the average cross-entropy [CDSU23]. When using our methodology on real data where the ground truth mutual information is unknown, a natural question is to assess the impact of an inaccurate estimation of the mutual information plugged into our bound. To show the impact of this limitation we added the inverse of the square root of the number of sample samples to the evaluation of the mutual information of each share. As a result, we obtain a region whose width shows the impact of an inaccurate mutual information evaluation. This illustrates that confidence intervals on the mutual information evaluation can be automatically translated into confidence intervals on the figures of merit with our bounds. As we will see on both datasets, this impact is mild on the final evaluation.

### 5.1 AES HD Dataset

We confirm our analysis with real traces from the AES HD\_ext dataset [BJP20]: <http://aisymlabdatasets.ewi.tudelft.nl/>. We targeted the 16 bytes of the first round of the AES. We checked using the SNR that the leakage of the first round occurs in the 200 first points of the traces, hence we restricted our attack window to these points. We profiled a *Regression Based Linear Discriminant Analysis* (RLDA) for each byte using SCALib framework [CB23] using the first 300.000 traces. We divided the remaining 200.000 traces into 40 subsets of 5.000 traces on which we performed  $N = 40$  independent attacks (one on each subset) using the profiled RLDAs and an increasing number of traces (ranging from 1 trace to 5.000 traces). We estimated the rank of the correct key for these attacks using the rank estimation technique based on histograms from SCALib [CB23] and plotted the results. We estimated the training information using the *RLDA Information Estimator*

implemented in SCALib to apply our bounds as reported in Table 1.

The results of the evaluations are given in Figure 6. The results are perfectly in line with the observations we made on synthetic data in Section 4.

Table 1: Estimated Informational Leakage on Each Byte.

| Index                                  | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12    | 13    | 14   | 15    |
|----------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|------|-------|
| S-box<br>(bits $\times$<br>$10^{-3}$ ) | 7.53 | 7.77 | 5.98 | 5.48 | 4.97 | 7.37 | 9.34 | 8.85 | 9.39 | 4.41 | 8.26 | 8.83 | 10.80 | 22.43 | 6.81 | 11.49 |

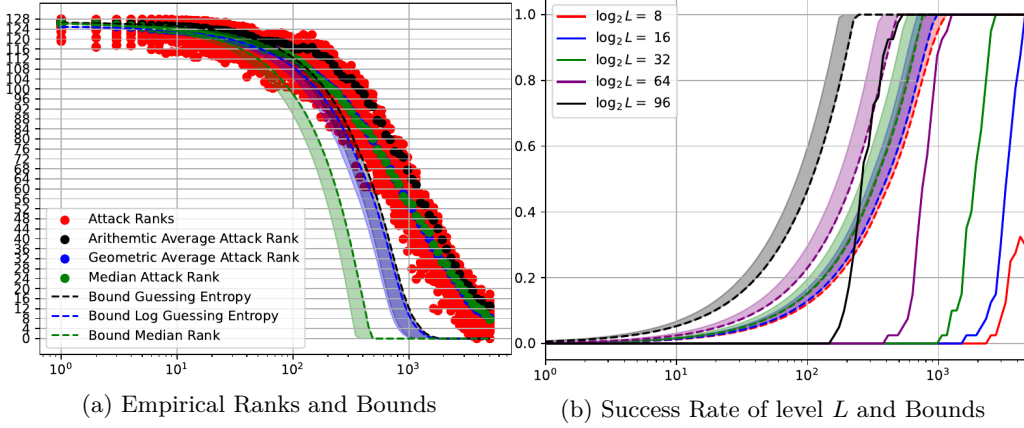


Figure 6: Results on AES-HD Dataset.

## 5.2 SMAesH Dataset

We also confirm our analysis with real traces from SMAesH dataset [CM24] available at <https://smaesh-challenge.simple-crypto.org/datasets.html>. The dataset is composed in three parts. A first part with variable keys is used for the profiling step and is composed of  $2^{24}$  traces. The uncompressed archive represents 141 GB of traces. The two other parts are made for attack evaluations with fixed keys  $fk0$  and  $fk1$  each composed with  $2^{24}$  traces. The uncompressed archives are both composed with 146 GB of traces. This is a realistic evaluation scenario with a target which is known to be hard to attack and the set of traces that are very long to acquire. Furthermore, the processing time for the attack evaluation is non-negligible and requires computational resources even with optimized libraries such as SCALib. We repeated  $N = \min\{\frac{2^{24}}{q}, 1000\}$  independent attacks with  $q$  traces for both  $fk0$  and  $fk1$ . This results into 1000 attacks with one trace but only two attacks with  $2^{23}$  traces. We could not perform more attacks since they are only  $2^{24}$  traces available for each fixed keys (which justifies our attack). For this reason, the empirical evaluation of the figures of merit on the right of the figure have a larger variance since they are estimated on few samples.

In this setting, the implementation is protected by state-of-the-art *Hardware Private Circuit* (HPC) masking scheme with two shares [CGLS21]. We consider the reference attack of the SMAesH challenge which is targeting the two shares of each byte at the output of the 16 S-boxes of the first round. This attack setting falls directly in the setting of our analysis. We can evaluate the informational leakage on each share as measured by mutual information and then use *Mrs. Gerber's Lemma* [BCG<sup>+</sup>23] to upper bound the informational leakage on the shared bytes. We can also directly evaluate the informational leakage on the unprotected key byte to avoid the small overhead due to Mrs. Gerber's Lemma. However, this second approach is harder to perform reliably as it requires to

evaluate a much smaller informational leakage (in the order of the square of informational leakage on the shares because of the masking countermeasure). The estimated leakage using both methodologies are reported in Table 2. In particular, both methodologies are consistent and indicates that the informational leakage on the unmasked S-boxes is in order of magnitude of  $10^{-6}$  bits per trace. The results are given in Figure 7 using the evaluation with Mrs. Gerber’s Lemma and shows the practicality of our approach for practical settings with state-of-the-art target that are hard to attack. We applied our bounds by keeping the  $i$ -th byte of the key that were leaking the least for  $i = 0, \dots, 15$  and kept the best bound. This slightly improves the bound because the different bytes of the key leak differently.

Table 2: Estimated informational leakage. Respectively on both shares, on the unmasked S-boxes using Mrs. Gerber’s Lemma and the unmasked S-boxes with direct evaluation.

| Byte Index                               | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   | 14   | 15   |
|------------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Share 0<br>(bits $\times 10^{-3}$ )      | 2.91 | 1.22 | 0.38 | 0.77 | 2.99 | 1.22 | 0.37 | 0.81 | 2.97 | 1.22 | 0.41 | 0.71 | 2.33 | 0.94 | 0.33 | 0.47 |
| Share 1<br>(bits $\times 10^{-3}$ )      | 1.24 | 3.72 | 0.54 | 0.77 | 0.97 | 3.71 | 0.51 | 0.75 | 1.03 | 3.73 | 0.53 | 0.70 | 0.71 | 2.81 | 0.33 | 0.36 |
| S-box MGL<br>(bits $\times 10^{-6}$ )    | 4.98 | 6.27 | 0.29 | 0.81 | 4.00 | 6.28 | 0.26 | 0.84 | 4.24 | 6.31 | 0.30 | 0.69 | 2.30 | 3.67 | 0.15 | 0.23 |
| S-box Direct<br>(bits $\times 10^{-6}$ ) | 4.68 | 4.86 | 1.29 | 1.41 | 4.37 | 5.06 | 1.00 | 1.20 | 4.93 | 4.85 | 1.07 | 1.10 | 3.48 | 3.65 | 0.71 | 0.57 |

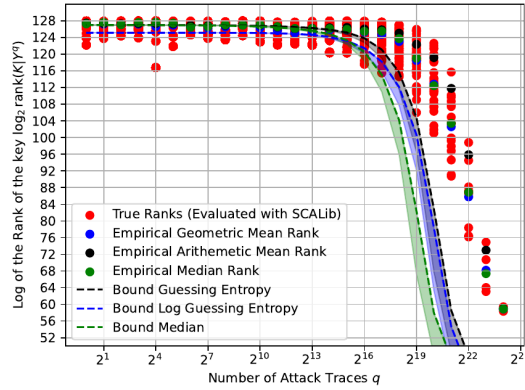


Figure 7: Results on SMAesH Dataset : Empirical Ranks and Bounds.

**Discussions** We observe on both the *AES HD* and *SMAesH* datasets that the observations made on simulated data from Section 4 extend to real traces. The *SMAesH* dataset is a protected implementation, harder to attack, and we confirm that our methodology scales to harder targets. Also, it shows that the methodology combines well with other shortcut formula, such as *Mrs. Gerbers Lemma* used to tightly upper bounds leakages in the presence of masking. Compared to the simulation setting, we observe on Figure 7 and Figure 6 a larger evaluation gap for the log-guessing entropy and median rank using our methodology. While our method is more conservative, the dashed curves are below the dotted values, in terms of evaluation of the number of traces required to mount the attack this only implies a factor 4 gap. It is possible that due to the masking countermeasure our bound is less tight and using *Mrs. Gerbers Lemma* also adds some gap in the estimation of mutual information. In terms of sensitivity to inaccurate mutual information estimation we confirm on Figure 7 and Figure 6 that the effect is mild on the final results. It only modifies

the evaluation of the number of traces to mount the attack by a moderate multiplicative factor. In our setting, we consider a divide-and-conquer setting on the 16-Sboxes after the first AddRoundKey. Hence, our bounds apply to this type of adversaries. The state-of-the-art attack on the SMAesH dataset, leverages more leakages in the diffusion layer of the AES to mount its attack and hence as expected outperforms our bound. Covering this type of attacks goes beyond the scope of this study.

## 6 Conclusion

We provided informational bounds on the figures of merit of side-channel attacks. Most notably we provided bounds on the success rate of level  $L$  of a side-channel attack bridging the important gap of taking into account the computational power of the adversary in security bounds based on informational metrics. As a byproduct, we obtained bounds on the  $\alpha$ -marginal guesswork and in particular a simple explicit bound for the median rank of the key. Furthermore, we obtained for the first time informational a bound on the geometric mean rank of the key (log-guessing entropy).

These formulas are of theoretical interest as they provide the dependency of the figure of merits with the number of traces (data complexity), the computational complexity and the level of noise in the traces.

Besides their theoretical interest the inequalities can be of practical interest for security evaluators. We confirm our findings on both synthetic data and real side-channel measurements. On the AES-HD dataset the full evaluation can be performed in a few minutes on a laptop. However, the evaluation on SMAesH dataset [CM24] had to be performed on a server with 96 processors (*AMD EPYC 7F72 24-Core Processor*) and 500 GB of random access memory. While the evaluation of mutual information on the full variable key dataset can be performed fast (approximately 56 minutes) even on a non-threaded implementation, the construction of the security graph takes several hours even when the attacks are parallelized in twelve different processes each leveraging multi-threading capability from SCALib. This evaluation process is still manageable on the SMAesH dataset. However, the largest benefit of our method is that building the security graph requires taking measurements with fixed keys and a large number of traces which can be extremely time-consuming. For SMAesH, only two fixed keys datasets are available because of the huge memory usage (150 GB per dataset) and time that would be required to acquire more traces. In particular, the security graph is based on only two different fixed keys while it should be averaged with many keys. Furthermore, the attacks with  $2^{24}$  traces can only be performed once for each key while they should be repeated. On the other hand, our methodology does not require obtaining measurements with fixed keys. Finally, we emphasize that if the masking order was 3 with the SMAesH implementation while maintaining a similar leakage per share then approximately  $10^3$  times more traces would be required to mount a successful attack according to our informational bounds. On the one hand, the evaluation of informational leakage would still be fast as its running time is proportional with the number of shares (because we rely on Mrs Gerbers Lemma as a shortcut formula for masked encodings). On the other hand, building the security graph would be extremely hard as the attack time (repeated  $100\times$  for a success rate estimation) would require  $100 \times 500$  hours assuming that the running time of the attack is linear in the number of traces and since the attack with  $2^{24}$  runs in 30 minutes with our implementation (i.e., 5.7 years if not performed in parallel). Not to mention the required  $1000 \times 21$  hours (i.e., 2.4 years) required for collection of the dataset that would be of 150 TB size (the SMAesH datasets took 21 hours each to be measured [CM24]). These orders of magnitude would be again multiplied with masking at order 4 with an even more intractable setting.

In terms of *Common Criteria Methodology* quotation system our results directly translates into (JIL rating [Joi20, Table 12]) the score for “elapsed time”. The score ranges

from 0 for an attack that can be performed in less than one hour to 10 points for an attack requiring more than four months to be performed. For SMAesH [CM24] each trace is measured in 4.46 ms, and we provide lower bounds on the rank of the correct key in the key ranking at the output of a maximum likelihood attack in terms of the number of traces used for the attack. Assuming that each key is enumerated in, say 1 ms, we obtain a score for the Common Criteria in terms of elapsed time as a sum of the two terms. Since we do not have to mount the attack our methodology offers a significant advantage for attacks that hard to mount.

Another practical usage of our lower bounds is that it can be used within fresh re-keying policy [AB00] to estimate when the ephemeral keys should be refreshed.

As an open question it would be interesting to obtain security guarantees when the adversary does not use its computational power to enumerate the key hypothesis but try to solve a cryptographically hard problem using the side-channel leakage as a trapdoor.



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